



HYPERTENSION – DEPI GRADUATION PROJECT

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DEPI - ROUND 3

■ **Executive Summary**

This project focuses on analyzing a real-world hypertension dataset to identify key factors contributing to high blood pressure and uncover meaningful health patterns. The workflow includes data collection, cleaning, exploratory data analysis (EDA), and the creation of interactive dashboards using Tableau and Power BI.

The analysis investigates the impact of age, stress levels, caffeine intake, sleep duration, smoking status, exercise level, and family history on hypertension prevalence. The study provides actionable insights to support public health decision-making and awareness programs. If a predictive model is included, its performance is evaluated using accuracy and other metrics to ensure reliability.

Overall, this project demonstrates the application of data analysis techniques to derive data-driven insights that can guide prevention strategies and improve healthcare outcomes.

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1. Project Planning & Management

1.1 Project Proposal:

■ Overview of the Project:

This project focuses on analyzing a real-world hypertension dataset to identify the major factors that contribute to high blood pressure. The goal is to apply data analysis techniques—including data cleaning, exploratory analysis, and data visualization—to extract meaningful insights that support public health decision-making. The project also aims to build a simple predictive model to demonstrate applied analytical skills.

■ Project Objective:

- Collect, clean, and preprocess a structured dataset.
- Explore relationships between variables such as age, stress, caffeine intake, sleep duration, and smoking.
- Create interactive dashboards using Tableau and Power BI.
- Provide insights and recommendations based on analytical findings.

■ Project Scope:

- Data collection from open datasets
- Data cleaning and preprocessing (handling missing values, duplicates).
- Exploratory Data Analysis (EDA).
- Visualization of key findings and business insights using Tableau and Power BI.
- Summarizing findings in a final report and presentation.

1.2 Project Plan:

■ Timeline:

Phase	Task	Duration	Output
Week 1	Data collection & understanding	3–5 days	Dataset ready
Week 2	Data cleaning & preprocessing	1 week	Clean dataset
Week 3	Exploratory Data Analysis (EDA)	1 week	Initial insights, Questions
Week 4	Visualization & Dashboarding	4–5 days	Tableau & Power BI
Week 6	Documentation & presentation	3 days	Final report

■ Key Milestones:

- Dataset collection and validation.
- Cleaning dataset & saving cleaned data to new file.
- EDA insights.
- Creating Dashboards using two different tools (Bowe BI, Tableau).
- Final Documentation.

■ Deliverables

- Cleaned and well-documented dataset.
- Power BI & Tableau Dashboard
- Project Documentation

■ Resource Allocation:

Software Tools: Python, Tableau, Power BI, Microsoft Excel, and word.

Time Allocation:

- 40% finding Data set & Data Cleaning: Handling missing values, duplicates, and formatting issues.
- 30% Exploratory Data Analysis (EDA): Identifying trends, correlations, and key insights.

- 20% Visualization: Designing dashboards in Tableau and Power BI.
- 10% Documentation: Preparing the final report.

1.3 Task Assignment & Roles:

- **Radwa Mohammed Ali – Hafsa Ahmed Abdullah:**
Power BI Dashboard - Data Cleaning
- **Omar Abdelazim Esmat – Youssef Hatem Saad Abouzeid:**
Tableau Dashboard 1 - Data Cleaning
- **Yasmin Adel Khalil – Shaza Khaled Abdelsalam:**
Tableau Dashboard 2 - Data Cleaning – Documentation

1.4 Risk Assessment & Mitigation Plan:

- **Risk 1:** Missing or inconsistent data
Mitigation: Apply cleaning methods using python.
- **Risk 3:** Misinterpretation of insights
Mitigation: Cross-check findings with mentor and multiple visualization techniques
- **Risk 4:** Time limitations
Mitigation: Follow weekly schedule strictly and prioritize core tasks.

2. Requirements Gathering

2.1 Stakeholders Analysis:

Stakeholder	Role	Responsibilities
Intern	Data Analyst Trainee	Perform data cleaning, analysis, and reporting
Data Analysis Mentor	Supervisor/Trainer	Provide guidance, feedback, and evaluation
Training Program Coordinator (CLS)	Internship Manager	Ensure project aligns with internship objectives
Company Management (DEPI)	Project Sponsor	Evaluate project outcome and impact
End Users	Data consumers	Use insights for Health Guidance

2.2 User Stories:

- As a data analyst, I want a clean dataset so I can perform accurate analysis.
- As a decision maker, I want dashboards to understand hypertension trends easily.
- As a researcher, I want to identify factors affecting hypertension.

2.3 Functional Requirements:

- Import and clean dataset
- Handle missing values & duplicates
- Perform EDA
- Build Tableau dashboard
- Build Power BI dashboard

3. Data Analysis & Workflow Design

3.1 Problem Statement:

Hypertension is a common chronic illness influenced by many lifestyle and genetic factors. The goal is to analyze the dataset to identify which factors contribute most to hypertension.

3.2 Data Source & Dictionary:

- **Data Source:** Kaggle Hypertension data set
- **Format:** hypertension.xlsx
- **Columns in the data set:**

Column Name	Data Type	Description
Age	Numerical	Age of the individual
Gender	Categorical	Male or Female
Sleep Duration	Numerical	Average hours of sleep per night
Stress Score	Numerical	Self-reported stress level (scale 1–10)
Caffeine Intake	Numerical	Daily caffeine consumption in milligrams
Salt Intake	Numerical	Estimated daily salt intake
Smoking Status	Categorical	Smoker or Non-Smoker
Exercise Level	Categorical	Physical activity level: Low, Moderate, High
Family History	Categorical	Whether the individual has a family history of hypertension (Yes/No)
Medication Type	Categorical	Type of medication used (e.g., Beta Blocker, ACE Inhibitor, Diuretic, Other)
Hypertension	Categorical	Diagnosis status (Yes/No)

3.3 Data Processing Workflow:

- import dataset
- Inspect data structure
- Handle missing values
- Detect and remove duplicates
- Fix data types
- Export cleaned dataset
- Perform EDA
- Build dashboards
- Documentation

4. Data Cleaning & Preprocessing

4.1 Tools Used:

- Python, pandas, NumPy,

4.2 Cleaning code

4.2.1 Overview Before Cleaning

```
[14]: import pandas as pd
import numpy as np

[15]: df = pd.read_excel("HyperTensionUNClean.xlsx")

df.head(5)
```

	Age	Salt_Intake	Stress_Score	BP_History	Sleep_Duration	BMI	Medication	Family_History	Exercise_Level	Smoking_Status	Has_Hypertension	Gender	Caffeine_Intake
0	25	8.5	0	Prehypertension	8.3	18.8	Diuretic	No	Low	Non-Smoker	Yes	Female	20
1	NaN	11.1	3.0	Prehypertension	7.2	NaN	Beta Blocker	Yes	High	Non-Smoker	No	Female	10
2	52.0	8.7	8	Normal	6.1	30.6	ACE Inhibitor	Yes	Moderate	Non Smoker	Yes	Male	40
3	84	8.2	0.0	Prehypertension	5.9	27.1	ACE Inhibitor	No	Low	Non-Smoker	Yes	Female	10
4	79.0	8	nan	Normal	7.6	23.7	Diuretic	Yes	Low	Non-Smoker	Yes	Male	20

4.2.2 Removing Spaces

Before:

```
[16]: print("BEFORE removing spaces:")
df.select_dtypes(include='object').head(5)
```

BEFORE removing spaces:

	Age	Salt_Intake	Stress_Score	BP_History	Sleep_Duration	BMI	Medication	Family_History	Exercise_Level	Smoking_Status	Has_Hypertension	Gender	Caffeine_Intake
5	25	8.5	0	Prehypertension	8.3	18.8	Diuretic	No	Low	Non-Smoker	Yes	Female	20
1	NaN	11.1	3.0	Prehypertension	7.2	NaN	Beta Blocker	Yes	High	Non-Smoker	No	Female	10
0	52.0	8.7	8	Normal	6.1	30.6	ACE Inhibitor	Yes	Moderate	Non Smoker	Yes	Male	40
4	84	8.2	0.0	Prehypertension	5.9	27.1	ACE Inhibitor	No	Low	Non-Smoker	Yes	Female	10
0	79.0	8	nan	Normal	7.6	23.7	Diuretic	Yes	Low	Non-Smoker	Yes	Male	20

After:

```
df_obj = df.select_dtypes(include='object').columns
for col in df_obj:
    df[col] = df[col].astype(str).str.strip()

print("AFTER removing spaces:")
df[df_obj].head(5)
```

AFTER removing spaces:

	Age	Salt_Intake	Stress_Score	BP_History	Sleep_Duration	BMI	Medication	Family_History	Exercise_Level	Smoking_Status	Has_Hypertension	Gender	Caffeine_Intake
0	25	8.5	0	Prehypertension	8.3	18.8	Diuretic	No	Low	Non-Smoker	Yes	Female	20
1	nan	11.1	3.0	Prehypertension	7.2	nan	Beta Blocker	Yes	High	Non-Smoker	No	Female	10
2	52.0	8.7	8	Normal	6.1	30.6	ACE Inhibitor	Yes	Moderate	Non Smoker	Yes	Male	40
3	84	8.2	0.0	Prehypertension	5.9	27.1	ACE Inhibitor	No	Low	Non-Smoker	Yes	Female	10
4	79.0	8	nan	Normal	7.6	23.7	Diuretic	Yes	Low	Non-Smoker	Yes	Male	20

4.2.3 Fixing Typos

Before:

```
[18]: print("BEFORE fixing typos:")
      df[['Has_Hypertension', 'Gender', 'Smoking_Status']].head(10)
```

BEFORE fixing typos:

```
[18]:
```

	Has_Hypertension	Gender	Smoking_Status
0	Yes	Female	Non-Smoker
1	No	Female	Non-Smoker
2	Yes	Male	Non Smoker
3	Yes	Female	Non-Smoker
4	Yes	Male	Non-Smoker
5	No	Female	Non-Smoker
6	No	Female	nan
7	Yes	Female	Non-Smoker
8	Yes	Male	Non-Smoker
9	Yes	Fem ale	Non-Smoker

After:

```
[19]: replace_map = {
      "Has_Hypertension": {
          "Yess": "Yes", "yes": "Yes", "YES": "Yes",
          "no": "No", "NO": "No", "N o": "No",
      },
      "Gender": {
          "male": "Male", "Male ": "Male", "M ale": "Male",
          "female": "Female", "Fem ale": "Female"
      },
      "Smoking_Status": {
          "smoker": "Smoker", "smoker": "Smoker", "Smokerr": "Smoker",
          "non smoker": "Non-Smoker", "NonSmokerr": "Non-Smoker"
      }
  }

  for col, mapping in replace_map.items():
      df[col] = df[col].replace(mapping)

  print("AFTER fixing typos:")
  df[['Has_Hypertension', 'Gender', 'Smoking_Status']].head(10)
```

AFTER fixing typos:

```
[19]:
```

	Has_Hypertension	Gender	Smoking_Status
0	Yes	Female	Non-Smoker
1	No	Female	Non-Smoker
2	Yes	Male	Non Smoker
3	Yes	Female	Non-Smoker
4	Yes	Male	Non-Smoker
5	No	Female	Non-Smoker
6	No	Female	nan
7	Yes	Female	Non-Smoker
8	Yes	Male	Non-Smoker

4.2.4 Fixing Numeric Types

Before:

```
[31]: print("BEFORE fixing numeric types:")
      df.dtypes
```

BEFORE fixing numeric types:

```
[31]: Age          object
      Salt_Intake   object
      Stress_Score  object
      BP_History    object
      Sleep_Duration object
      BMI           object
      Medication     object
      Family_History object
      Exercise_Level object
      Smoking_Status object
      Has_Hypertension object
      Gender         object
      Caffeine_Intake object
      dtype: object
```

After:

```
[100]: num_cols = ['Age', 'Salt_Intake', 'Stress_Score', 'Sleep_Duration', 'BMI', 'Caffeine_Intake']

      for col in num_cols:
          df[col] = pd.to_numeric(df[col], errors='coerce')
      df['Age'] = df['Age'].astype('Int64')

      print("AFTER fixing numeric types:")
      df.dtypes
```

AFTER fixing numeric types:

```
[100]: Age          Int64
      Salt_Intake   float64
      Stress_Score  float64
      BP_History    object
      Sleep_Duration float64
      BMI           float64
      Medication     object
      Family_History object
      Exercise_Level object
      Smoking_Status object
      Has_Hypertension object
      Gender         object
      Caffeine_Intake float64
      dtype: object
```

4.2.5 Handling Missing Values

Before:

```
[22]: print("Missing values BEFORE:")
      df.isna().sum()
```

Missing values BEFORE:

```
[22]: Age          61
      Salt_Intake    63
      Stress_Score   63
      BP_History     0
      Sleep_Duration 63
      BMI            64
      Medication     0
      Family_History 0
      Exercise_Level 0
      Smoking_Status 0
      Has_Hypertension 0
      Gender         0
      Caffeine_Intake 64
      dtype: int64
```

After:

```
[]: for col in df.columns:
      if df[col].dtype in ['float64', 'int64']:
          df[col] = df[col].fillna(df[col].median())
      else:
          df[col] = df[col].fillna(df[col].mode()[0])

      print("Missing values AFTER:")
      df.isna().sum()
```

Missing values AFTER:

```
[]: Age          0
      Salt_Intake    0
      Stress_Score   0
      BP_History     0
      Sleep_Duration 0
      BMI            0
      Medication     0
      Family_History 0
      Exercise_Level 0
      Smoking_Status 0
      Has_Hypertension 0
      Gender         0
      Caffeine_Intake 0
      dtype: int64
```

after

4.2.6 Handling Duplicates

```
[24]: print("BEFORE removing duplicates:")
      df.duplicated().sum()
```

BEFORE removing duplicates:

```
[24]: np.int64(83)
```

```
[25]: df = df.drop_duplicates()

      print("AFTER removing duplicates:")
      df.duplicated().sum()
```

AFTER removing duplicates:

```
[25]: np.int64(0)
```

4.2.7 Final Clean Data

```
[98]: print("FINAL CLEAN DATA SAMPLE:")
      df.head(10)
```

FINAL CLEAN DATA SAMPLE:

```
[98]:
```

	Age	Salt_Intake	Stress_Score	BP_History	Sleep_Duration	BMI	Medication	Family_History	Exercise_Level	Smoking_Status	Has_Hypertension	Gender	Caf
0	25	8.5	0.0	Prehypertension	8.3	18.8	Diuretic	No	Low	Non-Smoker	Yes	Female	
1	79	11.1	3.0	Prehypertension	7.2	26.0	Beta Blocker	Yes	High	Non-Smoker	No	Female	
2	52	8.7	8.0	Normal	6.1	30.6	ACE Inhibitor	Yes	Moderate	Non Smoker	Yes	Male	
3	84	8.2	0.0	Prehypertension	5.9	27.1	ACE Inhibitor	No	Low	Non-Smoker	Yes	Female	
4	79	8.0	5.0	Normal	7.6	23.7	Diuretic	Yes	Low	Non-Smoker	Yes	Male	
5	18	6.8	10.0	Normal	6.0	23.8	Beta Blocker	Yes	Low	Non-Smoker	No	Female	
6	21	11.6	9.0	Normal	6.9	26.0	Beta Blocker	No	High	nan	No	Female	
7	45	9.7	10.0	Hypertension	9.6	26.3	nan	Yes	Low	Non-Smoker	Yes	Female	
8	53	3.0	5.0	Normal	7.5	22.2	ACE Inhibitor	Yes	Moderate	Non-Smoker	Yes	Male	
9	50	8.9	8.0	Hypertension	3.9	33.9	ACE Inhibitor	Yes	Low	Non-Smoker	Yes	Female	

5. Exploratory Data Analysis (EDA)

After cleaning the dataset, exploratory analysis was conducted using Tableau and Power BI to uncover patterns and correlations related to hypertension. The dashboards provided interactive visualizations that helped identify key health factors influencing hypertension risk.

5.1 Analytical Questions Explored:

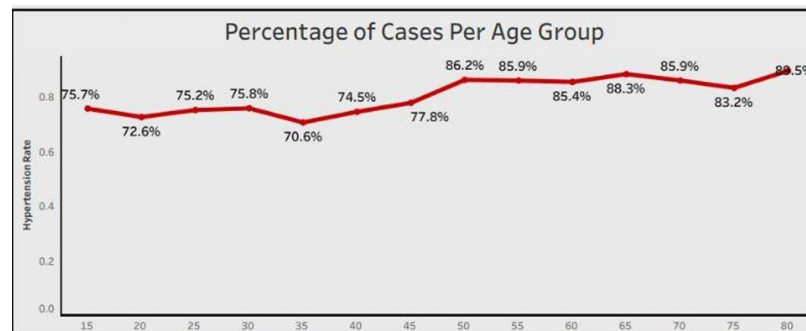
- How does age affect the likelihood of developing hypertension?
- Is there a correlation between stress levels and hypertension?
- Does caffeine intake influence hypertension risk?
- How does sleep duration relate to hypertension cases?
- Are females more likely to have hypertension than males?
- What is the impact of family history on hypertension diagnosis?
- Which type of medication is most used by hypertensive patients?
- How does smoking status relates to exercise level among hypertensive individuals?
- Is obesity (high BMI) associated with hypertension?
- How does exercise level affect hypertension prevalence?
- What is the distribution of medication types among hypertensive individuals?
- How does blood pressure history relate to medication usage?
- What are the average stress levels across different categories?
- How does BMI vary across weight categories?
- What is the relationship between family history and hypertension?

5.2 Insights From Tableau Dashboards:

5.2.1 Age vs Hypertension Rate

Insight: Hypertension prevalence increases significantly with age, especially after 40.

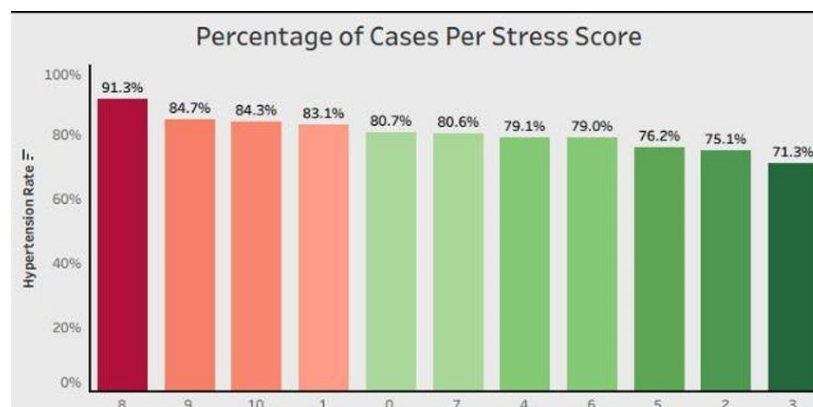
Evidence: Line chart shows rates rising from 72.6% at age 15 to 89.5% at age



5.2.2 Stress Score vs Hypertension Rate

Insight: Individuals with stress scores of 8–10 show hypertension rates above 84%, peaking at 91.3%.

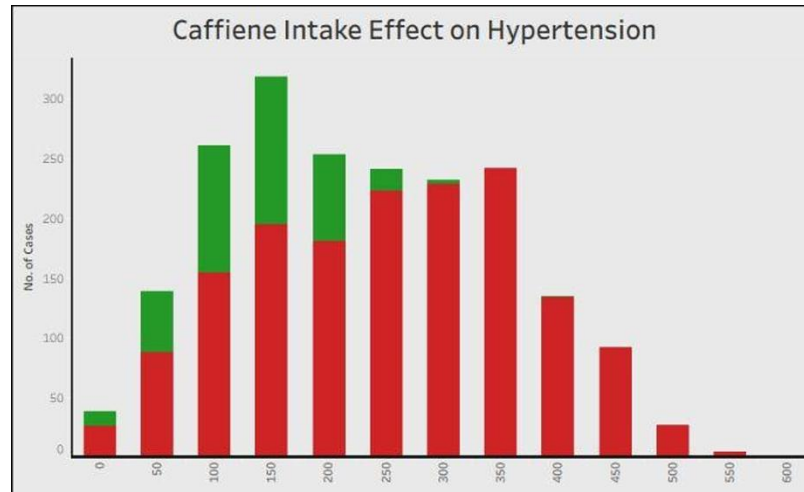
Evidence: Bar chart visualizes the strong positive correlation.



5.2.3 Caffeine Intake Effect

Insight: Higher caffeine consumption is associated with increased hypertension cases.

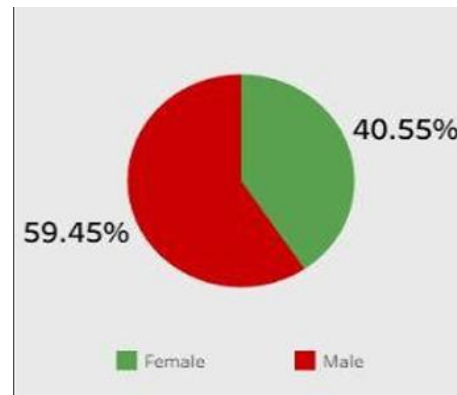
Evidence: Bar chart shows red bars (hypertension) rising with caffeine levels.



5.2.4 Gender Distribution

Insight: Females represent a higher proportion of hypertension cases.

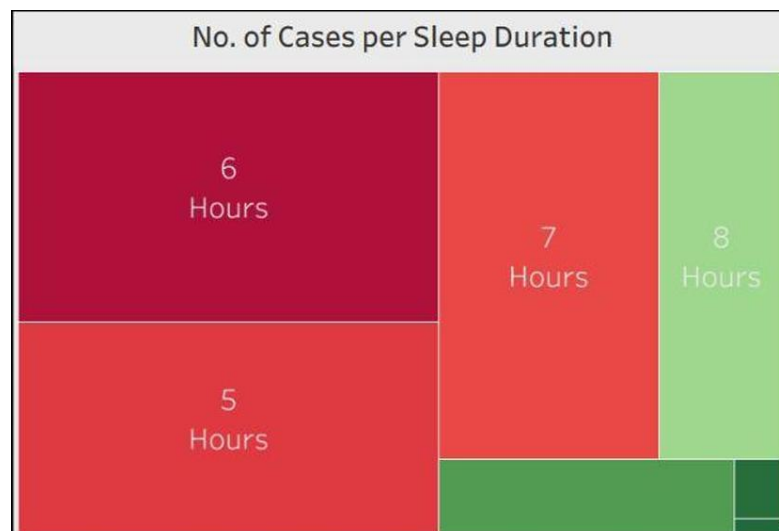
Evidence: Pie chart shows 59.45% female vs 40.55% male



5.2.5 Sleep Duration

Insight: Most hypertension cases are concentrated among individuals sleeping 5–6 hours.

Evidence: Tree map shows distribution by sleep duration.



5.2.6 Obesity & Hypertension

Insight: Individuals with higher BMI values are more likely to have hypertension.

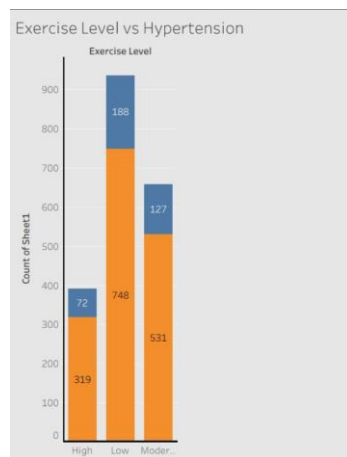
Evidence: Scatter plot shows orange dots (hypertensive) clustered in the obese BMI range.



5.2.7 Exercise Level vs Hypertension

Insight: Lower exercise levels are associated with higher hypertension rates.

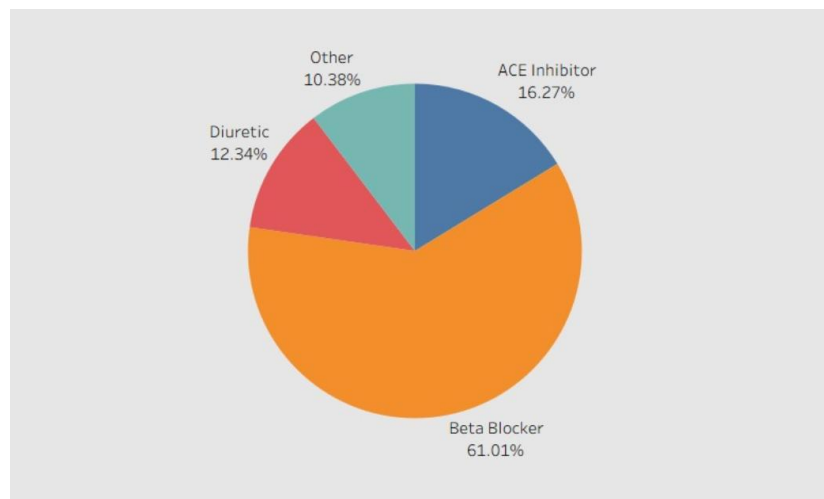
Evidence: Bar chart shows 748 hypertensive cases in the low exercise group, compared to 319 (high) and 531 (moderate).



5.2.8 Medication Distribution

Insight: Beta Blockers are the most prescribed medication for hypertension.

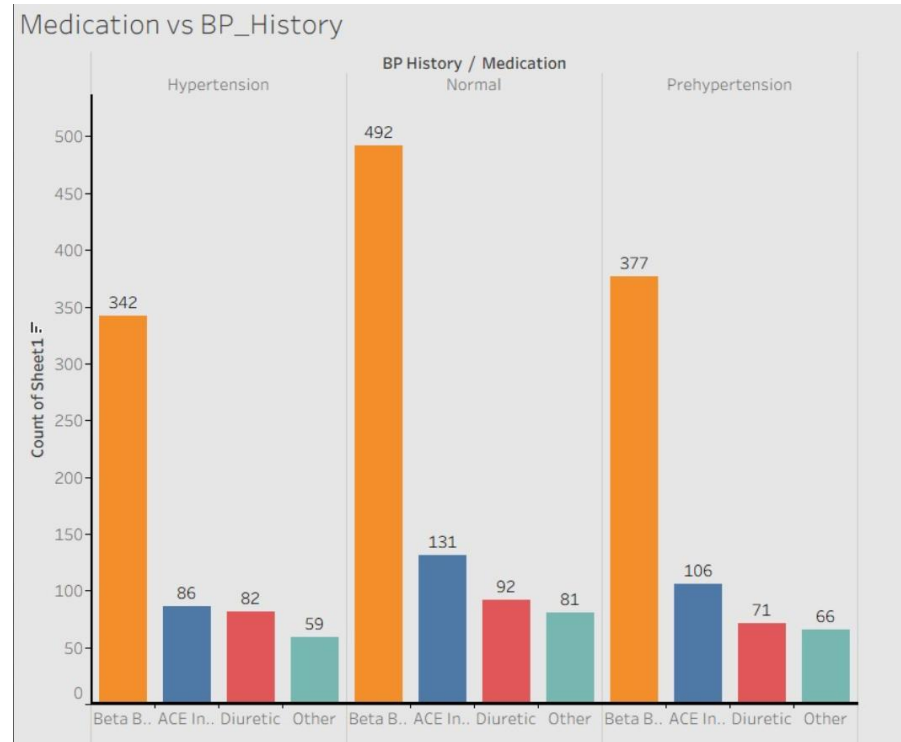
Evidence: Pie chart shows Beta Blockers at 61.01%, followed by ACE Inhibitors (16.27%) and Diuretics (12.34%).



5.2.9 BP History & Medication Use

Insight: Individuals with a history of hypertension are more likely to be prescribed Beta Blockers.

Evidence: Bar chart shows 342 Beta Blocker cases among hypertensive patients, compared to 492 (normal) and 377 (prehypertension).



5.2.10 Comparison of Health Factors By Hypertension Status

Insight: Individuals with **low exercise levels**, a **family history of hypertension**, and who are **smokers** show the **highest percentage of hypertension cases** compared to other combinations of health factors.

Evidence: The table highlights that when these three risk factors are combined, the proportion of individuals with hypertension is significantly higher than those without.

Comparison of Health Factors by Hypertension Status							
Exercise ..		Family Hi..	Smoking St..	Has Hypertension			
				No		Yes	
High	No	Non-Smoker		▼	27.50%	👤	72.50%
		Smoker		▼	14.63%	👤	85.37%
	Yes	Non-Smoker		▼	17.90%	👤	82.10%
		Smoker		▼	5.88%	👤	94.12%
Low	No	Non-Smoker		▼	29.00%	👤	71.00%
		Smoker		▼	19.01%	👤	80.99%
	Yes	Non-Smoker		▼	17.87%	👤	82.13%
		Smoker		▼	7.86%	👤	92.14%
Moderate	No	Non-Smoker		▼	28.79%	👤	71.21%
		Smoker		▼	20.51%	👤	79.49%
	Yes	Non-Smoker		▼	17.56%	👤	82.44%
		Smoker		▼	6.67%	👤	93.33%
Grand Total				▼	19.50%	👤	80.50%

5.2.11 Stress Level Averages

Insight: Average stress levels vary significantly across categories.

Evidence: High stress: 8.509, Medium: 5.058, Low: 1.481.

StressLevel Avg	
High	8.509
Low	1.481
Medium	5.058

5.2.12 BMI by Weight Category

Insight: BMI values align with expected weight classifications.

Evidence: Obese: 32.55, Overweight: 27.31, Normal: 22.37, Underweight: 16.60.

Average BMI	
Normal	22.37
Obese	32.55
Overweight	27.31
Underweight	16.60

5.2.13 Family History Count

Insight: A large portion of the dataset includes individuals with a family history of hypertension.

Evidence: Count of FH: 1,985 cases.

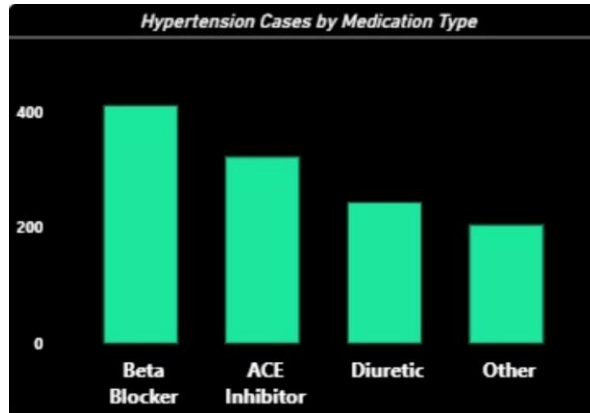
Count of FH
1,985

5.3 Insights From Bower BI Dashboards:

5.3.1 Medication Type and Hypertension

Insight: Beta Blockers are the most common medication among hypertensive patients.

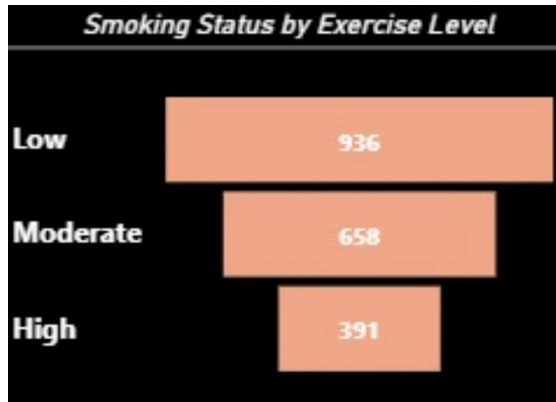
Evidence: Bar chart shows ~400 cases for Beta Blockers, followed by ACE Inhibitors and Diuretics.



5.3.2 Smoking Status vs Exercise Level

Insight: Smokers tend to have lower exercise levels.

Evidence: Horizontal bar chart shows 935 cases in the low exercise group



5.3.3 Caffeine Intake by Stress Level

Insight: Higher stress levels are linked to increased caffeine consumption.

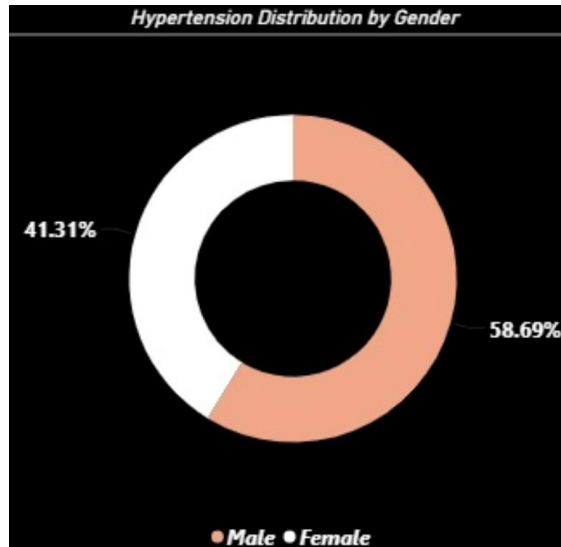
Evidence: Waterfall chart shows caffeine intake increasing with stress scores from 4 to 10.



5.3.4 Gender and Hypertension

Insight: Females have a higher hypertension rate than males.

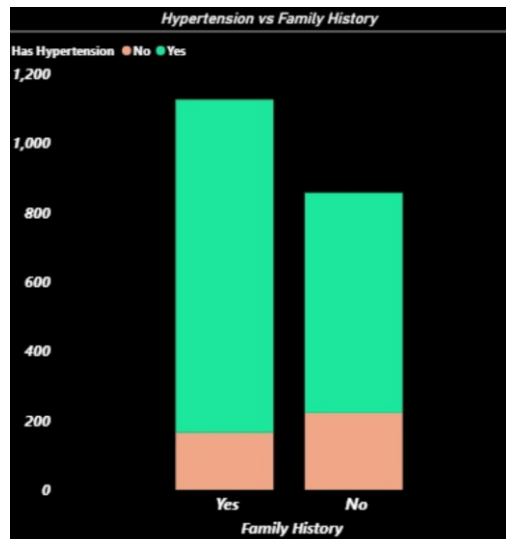
Evidence: Donut chart shows 58.69% female vs 41.31% male.



5.3.5 Family History Impact

Insight: Individuals with a family history of hypertension are more likely to be diagnosed.

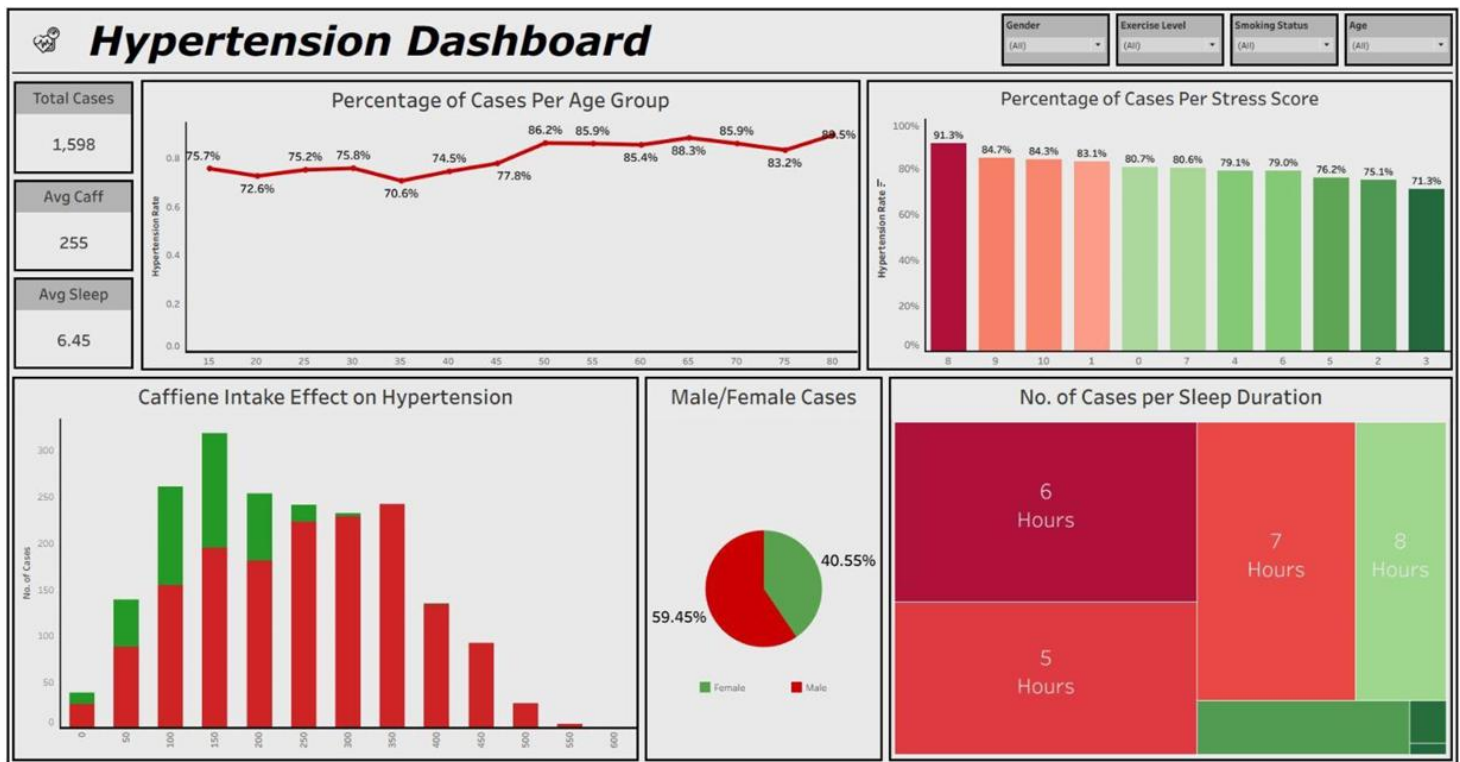
Evidence: Bar chart shows ~1000 hypertensive cases among those with family history.



6. Visualization & Dashboard Overview

Three dashboards were created using **Tableau** and **Power BI** to present the findings from the hypertension dataset in a clear, interactive format. Each dashboard focuses on different aspects of the data (lifestyle factors, medical history, and behavioral patterns), and helps communicate insights to both technical and non-technical audiences.

6.1 Dashboard 1 – Hypertension Overview (Tableau)



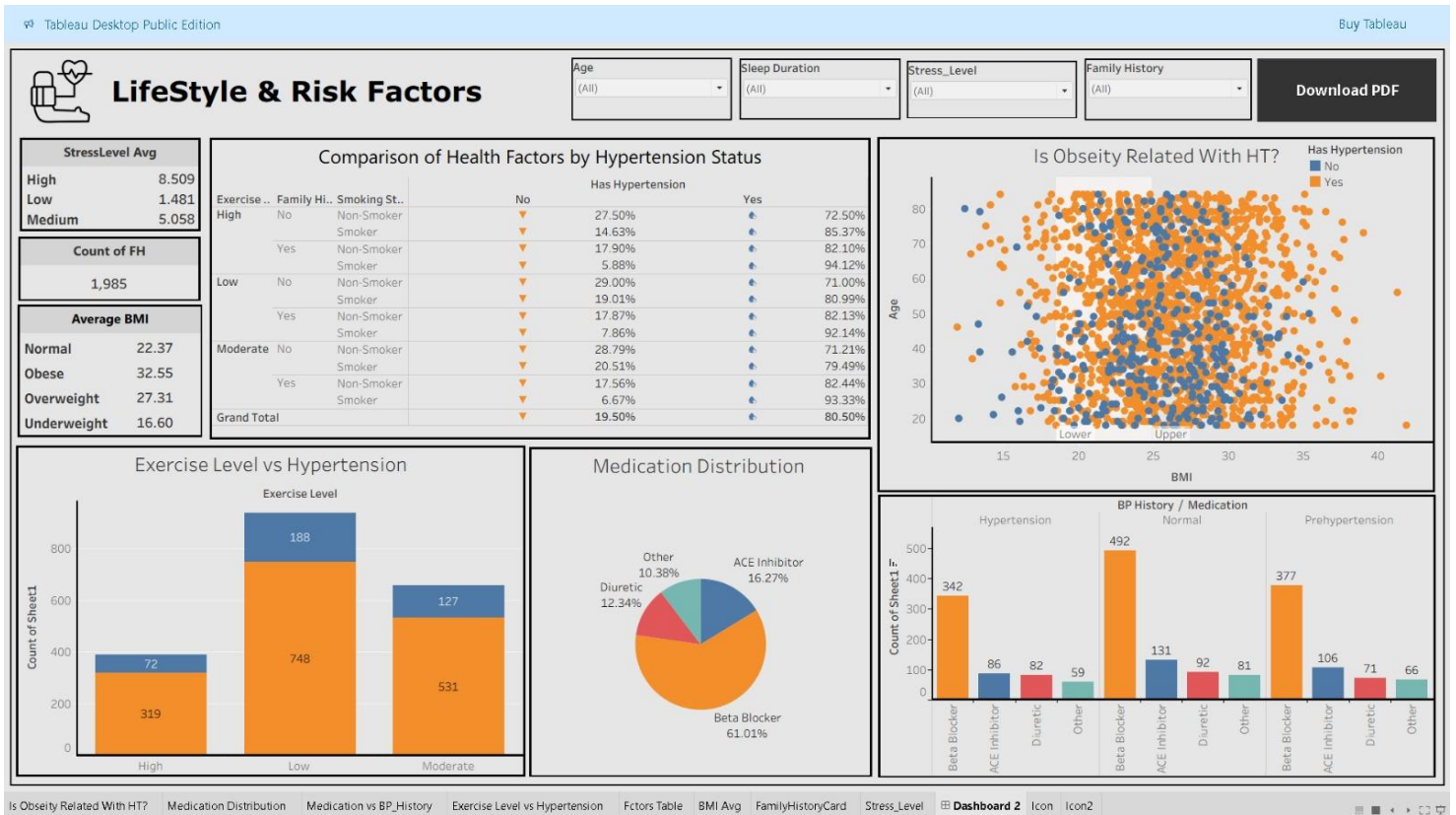
■ **Purpose:** To explore how age, stress level, caffeine intake, sleep duration, and gender relate to hypertension.

■ **Design Highlights:**

- **Line chart:** Percentage of cases per age group
- **Bar chart:** Percentage of cases per stress score
- **Pie chart:** Gender distribution
- **Treemap:** Sleep duration distribution
- **Bar chart:** Caffeine intake effect on hypertension
- **Cards:** Total Cases, Avg Caff, Avg sleep

■ **Interactivity:** Includes filters for gender, exercise level, smoking status, and age.

6.2 Dashboard 2 – Lifestyle & Risk Factors (Tableau)



■ **Purpose:** To examine how BMI, exercise level, smoking, and family history interact with hypertension risk.

■ **Design Highlights:**

- Scatter plot: Obesity vs hypertension status
- Bar chart: Exercise level vs hypertension
- Pie chart: Medication distribution
- Bar chart: BP history vs medication type
- Table: Comparison of health factors by hypertension status

■ **Interactivity:** Includes filters for age, sleep duration, stress level, and family history.

6.3 Dashboard 1 – Medical & Behavioral Factors (Bower BI)



- **Purpose:** To analyze the impact of medication type, smoking status, exercise level, and family history on hypertension.
- **Design Highlights:**
 - **Bar chart:** Hypertension cases by medication type
 - **Horizontal bar chart:** Smoking status vs exercise level
 - **Waterfall chart:** Caffeine intake pattern by stress level
 - **Donut chart:** Hypertension distribution by gender
 - **Bar chart:** Hypertension vs family history
- **Interactivity:** Includes filters for hypertension status and smoking status.

7. Conclusion

The analysis of the Hypertension dataset, supported by Python preprocessing and visual exploration in Tableau and Power BI, revealed several important patterns:

- 1- **Age:** Hypertension prevalence increases significantly with age, especially after 40.
- 2- **Stress:** Higher stress scores (8–10) are strongly correlated with hypertension.
- 3- **Caffeine Intake:** Excessive caffeine consumption is associated with higher hypertension cases.
- 4- **Sleep Duration:** Shorter sleep (5–6 hours) is linked to increased hypertension prevalence.
- 5- **Gender:** Females represent a slightly higher proportion of hypertension cases compared to males.
- 6- **Family History:** Individuals with a family history of hypertension are more likely to be diagnosed.
- 7- **Lifestyle Factors:** Smokers tend to have lower exercise levels, further increasing risk.
- 8- **Medication:** Beta Blockers are the most common medication among hypertensive patients.

These findings highlight the multifactorial nature of hypertension, influenced by lifestyle, genetics, and medical treatment.

8. Recommendations

Based on the insights, the following recommendations can be made:

Promote Healthy Lifestyle Choices

- Encourage regular physical activity to reduce risk.
- Advocate for reduced caffeine and salt intake.

Stress Management Programs

- Implement workplace and community initiatives to help individuals manage stress effectively.

Sleep Hygiene Awareness

- Educate individuals on the importance of adequate sleep (7–8 hours) for cardiovascular health.

Targeted Screening

- Prioritize hypertension screening for individuals over 40 and those with a family history.

Gender-Specific Health Campaigns

- Address the slightly higher prevalence among females with tailored awareness programs.

Medication Monitoring

- Ensure patients on Beta Blockers and other medications are monitored regularly for effectiveness and side effects.

The combination of **Python preprocessing** and **BI, Tableau dashboards** provided a comprehensive view of hypertension risk factors. This approach demonstrates how data-driven insights can guide public health strategies, clinical decision-making, and individual lifestyle improvements.